3.2

Detecting Equal Values with Branches

Using an if statement and the equality operator ==



When does a program need equality?

A decision often begins by asking: “Are these two values the same?”



Hotel discount example

A hotel gives a discount to guests celebrating their 50th wedding anniversary.

Question for the program:

Is numYears equal to 50?

If yes → apply the anniversary discount

The if statement

An if statement executes a group of statements only when its expression is true.

```
if (condition) {
    // statements execute if
condition is true
}
```

Key idea

true → enter the branch
false → skip the branch



Using == to test equality

The equality operator compares the left and right sides.

```
numYears == 50
```

If numYears contains 50, the expression evaluates to:

true



== asks a question: “Are these equal?”

Complete example: 50th anniversary discount

```
int numYears;

numYears = scnr.nextInt();

if (numYears == 50) {
   System.out.println("Anniversary
discount!");
}
```

1 Compare

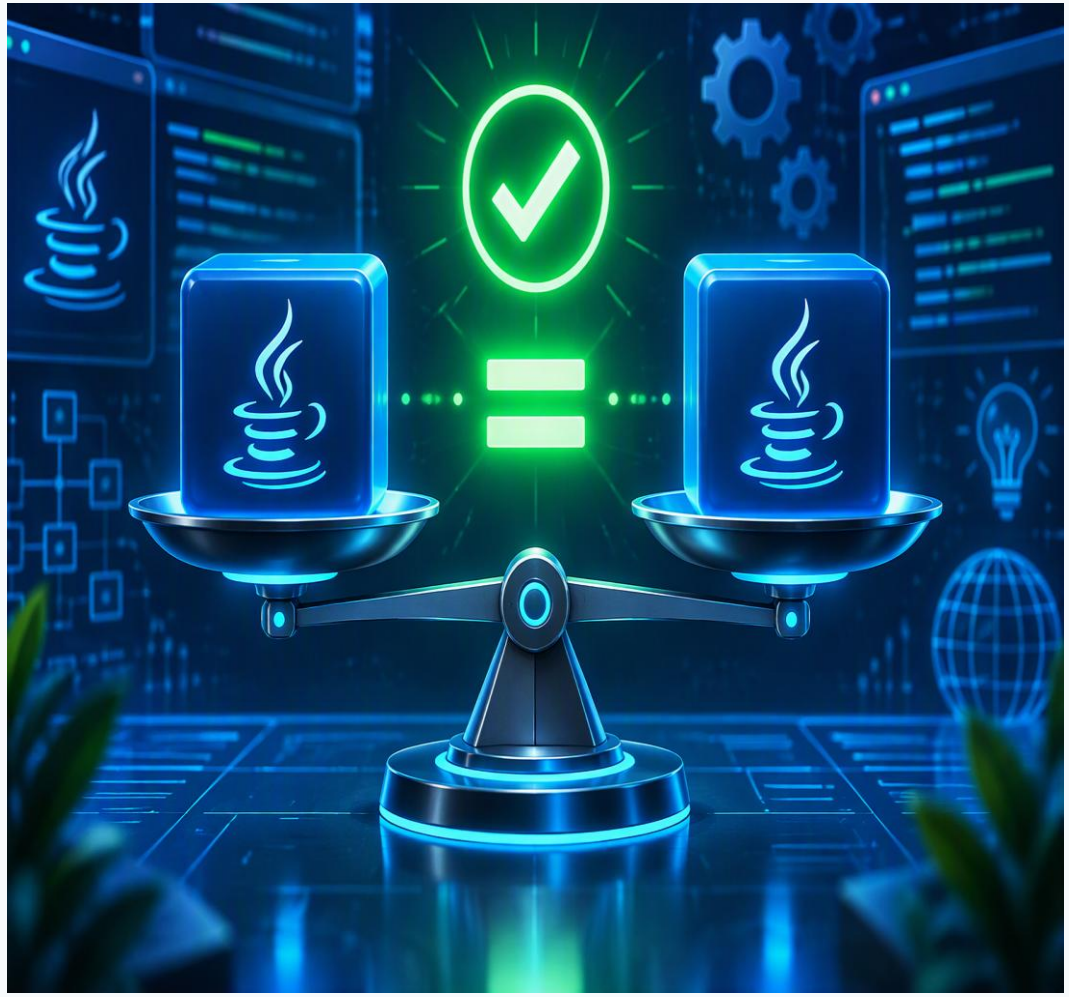
`numYears == 50`

↓

2 If true, execute the statements in { }

Important: == is not =

A common beginner mistake is confusing comparison with assignment.



`==` Equality operator

Compares two values → true or false

`=` Assignment operator

Stores a value in a variable

Braces group the branch statements

Braces { } surround the statements controlled by the if.

```
if (numYears == 50) {
    System.out.println("Discount applied");
    System.out.println("Congratulations!");
}
```

{ } braces

group statements

[ ] brackets

different symbols

Use the correct name: { } are braces; [ ] are brackets.

Good practice: indent branch statements

Indentation makes the branch easy to see and read.

Recommended style

```
if (numYears == 50) {
   System.out.println("Discount!");
}
```

zyBooks uses 3 spaces for indentation.

Harder to read

```
if (numYears == 50) {
System.out.println("Discount!");
}
```

Be consistent throughout your program.

Quick check

Predict the result before running the code.

Example A

```
int numYears = 50;
if (numYears == 50) {
   System.out.println("Discount");
}
```

Does the branch execute? YES

Example B

```
int numYears = 25;
if (numYears == 50) {
   System.out.println("Discount");
}
```

Does the branch execute? NO

3.2 Summary

Detecting equal values with branches

- Use an if statement to execute statements when a condition is true.
- Use == to test whether two values are equal.
- **Do not confuse == (comparison) with = (assignment).**
- **Use braces { } to group the statements in the if branch.**
- Indent branch statements consistently; zyBooks uses 3 spaces.

